

Solution Requirements

Date	08 October 2026
Team ID	UNM1611GUFRAN/SWTID-2026-9802
Project Name	AI FAQ Assistant API
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

We defined functional requirements like secure login, FAQ CRUD, AI answers, and search to specify what the system will do.

Non-functional requirements ensure speed, scalability, security, and reliability, describing how the system will perform.

- **Functional Requirements (FR)** defines specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specifies the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Secure login with JWT</i>	<i>High</i>
2	Authorization levels	<i>Role-based access (user/admin)</i>	<i>High</i>
3	External interfaces	<i>Google Gemini API</i>	<i>High</i>

		<i>integration</i>	
4	Transactions processing	<i>CRUD operation on FAQs</i>	<i>High</i>
5	Reporting	<i>Search results, structured FAQs</i>	<i>Medium</i>
6	Business rules, etc.	<i>Category validation (Technology, Education, Health, Banking, General)</i>	<i>High</i>
7	Compliance to laws or regulations	<i>Secure password hashing, JWT standards</i>	<i>High</i>
8	<i>Other</i>	<i>Nil</i>	<i>Nil</i>

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>[Describes response times, processing speed, etc.]</i>	<i>API response < 2 seconds</i>
2	Scalability	<i>[Describes how the system handles growth in users or data]</i>	<i>10,000 + FAQs</i>
3	Security & Data Privacy	<i>[Detailed data protection, encryption, and vulnerability management]</i>	<i>AES-256 encryption</i>
4	Reliability & Availability	<i>Specifies uptime requirements and fault tolerance</i>	<i>99.9% uptime</i>
5	Usability & Accessibility	<i>Describes user experience standards and accessibility compliance</i>	<i>RESTful endpoints, Postman collection included</i>
6	Other	<i>[Add any additional non-functional requirements (e.g., Maintainability, Portability)]</i>	<i>Nil</i>